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Hybrid quantum annealing for large-scale exam scheduling: Validation in real-world educational scenarios☆

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HIGHLIGHTS

- Quantum annealing solves 2749 exams with zero student conflicts in practice.
- Transforms exam constraints into quantum-solvable quadratic optimization models.
- Integrates D-Wave quantum hardware for efficient optimization of complex models.
- Graph-based preprocessing reduces QUBO complexity and qubit usage.
- Validates quantum computing's potential in educational resource allocation.

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ABSTRACT

This study applied the hybrid quantum annealing to address complex exam scheduling challenges in large-scale educational scenarios through hybrid quantum annealing enhanced by graph-based preprocessing. By formulating the problem as a Quadratic Unconstrained Binary Optimization (QUBO) model and leveraging the D-Wave Advantage system, our method integrates quantum annealing with classical preprocessing to resolve constraints on exam room availability, student-course conflicts, and batch synchronization. The approach was applied in a university's 2022 pandemic-era makeup exam scheduling for 1807 students and 215 courses (2749 exam instances) with zero conflicts. Experimental results show that hybrid quantum annealing consumes merely 86 ms of Quantum Processing Unit (QPU) execution time. In contrast, classical simulated annealing requires 13547 ms of Central Processing Unit (CPU) execution time for the same problem scale. This work bridges quantum computing and educational operations, offering a comparative analysis of hybrid algorithms in multi-constraint optimization domains.

1. Introduction

Exam scheduling is one of the important tasks the academic affairs office in universities must deal with every semester [1]. Conflicts are less likely to occur for end-of-term exams due to ample scheduling time and the availability of multiple classrooms. Typically, with the coordination of academic affairs personnel, things can be smoothly arranged. However, for special exams such as makeup exams, retake exams, and graduation clearance exams, there are often requirements to complete them within a few days. These exams involve a larger number of courses, and each student has a unique set of courses, leading to frequent conflicts in exam scheduling. Additionally, to avoid disrupting regular

☆ Z.Z. conceived the study, developed the methodology, conducted the experiments, analyzed the results, and wrote the original draft. Q.C. assisted with data collection and preprocessing. C.Z. advises on fundamental research in quantum computing. M.T. verifies the algorithm proposed in this paper in practical teaching activities. S.L. supervised the study, acquired funding, provided critical feedback, and edited the manuscript. All authors have read and approved the final manuscript.

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classes, the number of available classrooms is limited for scheduling exams, posing significant challenges to the scheduling capabilities of the administrative office.

In these special exams, to prevent exam paper leaks, the exam times for the same course must be synchronized. Moreover, a student cannot take exams for multiple courses simultaneously, which necessitates arranging different courses for each student at different times. These constraints require teachers to perform multiple checks, a process that is both intricate and prone to errors. Consequently, there is a pressing need for an efficient exam scheduling algorithm to assist the administrative office in its tasks.

Currently, common exam scheduling algorithms include graph colouring algorithm (GCA) [2–5], ant colony optimization (ACO) [6,7], artificial bee colony algorithm (ABC) [8], discrete particle swarm optimization (DPSO) [9], genetic algorithm (GA) [10,11], and simulated annealing algorithm (SA) [12–15]. Recently, with the development of quantum computing, the quantum annealing (QA) algorithm has shown powerful global search ability in solving combinatorial optimization problems [16,17]. Constrained by the limitations of classical computers, the quantum annealing algorithm running on a quantum computer demonstrates its exceptional performance [18,19]. Therefore, this paper plans to use the quantum annealing algorithm to solve complex exam scheduling problems for better solutions.

This paper focuses on the special complex exams, such as makeup exams, retake exams, and graduation clearance exams, rather than end-of-term exams. The exam scheduling problem is transformed into a Quadratic Unconstrained Binary Optimization (QUBO) model by converting objective functions and constraints to a Hamiltonian. This model is processed by the D-Wave quantum computer, where a global search is performed through qubit coupling adjustments to find the optimal exam arrangement that minimizes system energy.

This research explores a novel solution approach for complex exam scheduling and enriches quantum annealing applications.

2. Related works

Quantum annealing has demonstrated promise in scheduling domains but faces scalability challenges in real-world deployments [20]. Early studies on nurse scheduling (NSP) using D-Wave 2000Q achieved limited success with fixed schedules, and the realization of quantum annealers is insufficient in capacity to explore realistic sizes [21]. Recent industrial applications highlight QA's potential in multi-constraint scenarios:

Transport Networks: Dixit et al. [22] formulated the transport network design problem (TNDP) as a bi-level problem, with the upper level modeled as a QUBO problem. They solved it using quantum annealing on a D-Wave quantum computer. They compared the quantum annealing results with tabu search. The results showed that quantum annealing provides significant computational speedup over the classical tabu search method. This demonstrates the potential of quantum annealing in solving practical network design optimization problems across different contexts, including traffic. However, for small networks (<150 links), the advantage of quantum annealing speed is not significant, and pure quantum methods cannot be directly applied.

Finance: Xu et al. [23] designed a hybrid quantum-classical algorithm for dynamic asset allocation to meet target return and risk limits. They modeled the Markowitz portfolio optimization problem as a quadratic unconstrained binary optimization problem. Experimental results showed that the quantum solutions can achieve up to 80 % of the classical optimum's return while satisfying risk constraints. This proves the immense potential of hybrid algorithms in quantum-enhanced portfolio optimization. The performance of this method is limited by its topology and chain length, and large-scale problems still require classical post-processing.

Power System: Paterakis et al. [24] proposed a hybrid quantum-classical algorithm to solve general mixed-integer linear programming problems and applied it to unit commitment in power systems. They significantly reduced the number of iterations by accelerating Benders decomposition through a multi-cut strategy and quantum computing, and verified the feasibility and effectiveness of the method through experiments. This highlights the promising applications of hybrid algorithms in solving optimization problems in critical areas. However, limited by quantum hardware, experiments are only validated on small-scale systems (such as 8-bus and 30-bus), and solving large-scale problems still relies on classical computing.

Automated tools: Tosun et al. [25] transformed the quadratic assignment problem (QAP) instance into a QUBO model, providing new ideas for the application of quantum computing in combinatorial optimization problems. The experiment only validated the precise transformation on small-scale instances ($n \leq 20$), and its applicability to larger-scale problems was not fully demonstrated. Furthermore, the comparison of quantum computing and traditional methods in terms of efficiency was not thoroughly discussed.

Job shop scheduling problem: Carugno et al. [20] demonstrated the potential of D-Wave quantum annealers for solving small-scale Job Shop Scheduling Problems (JSSP) with quantum annealing outperforming classical simulated annealing in specific cases and reverse annealing enhancing solutions. However, the approach faces critical limitations: converting large JSSP instances (size=26) to QUBO models requires over 8 h of computation and introduces numerous auxiliary variables; hardware topology constraints (Chimera/Pegasus) necessitate qubit chains, leading to thousands of physical qubits per instance and a 37 % chain break rate that compromises solution reliability. These challenges highlight the inferior scalability/stability compared to classical tabu search in larger or unknown-optima scenarios, urging algorithm-hardware co-optimization for progress.

Previous studies have demonstrated the effectiveness of quantum annealing in transportation, finance, power systems, and industrial scheduling, but have rarely focused on real-world educational scenarios. Exam scheduling in academic institutions involves multi-constraint optimization with distinctive structural features, yet remains underexplored in the context of quantum computing. This study applies a hybrid quantum-classical algorithm to tackle complex exam scheduling by constructing a QUBO model. The method is validated through a large-scale educational case, showing that quantum annealing can be effectively applied to the educational domain and offering new insights into its broader applicability.

3. Theoretical background

Quantum annealing has emerged as a prominent approach to solving complex combinatorial optimization problems. Rooted in the principles of adiabatic quantum computing, this technique encodes optimization objectives into Ising models or equivalent QUBO formulations. Unlike classical simulated annealing, which relies on thermal fluctuations to escape the local minima, quantum annealing leverages quantum tunneling and superposition to traverse energy barriers more efficiently, enabling simultaneous exploration of vast solution spaces [26]. This capability positions it as a transformative tool for domains such as logistics, finance, and scheduling, where exponential solution spaces defy traditional algorithms. The hybrid quantum-classical approaches mitigate hardware limitations while demonstrating tangible advantages in specific applications where quantum annealing accelerates solution discovery and enhances solution quality compared to classical heuristics.

3.1. Quantum annealing

Quantum annealing is an emerging quantum computing technique with significant potential for solving hard optimization problems. It is implemented on specialized quantum hardware known as a Quantum Processing Unit (QPU), which processes quantum bits (qubits) instead of

classical binary bits. Unlike traditional Central Processing Unit (CPU) that executes operations sequentially, QPU harnesses quantum mechanical principles such as superposition and quantum tunneling, allowing for the parallel exploration of multiple solutions within a vast search space. This enhances the likelihood of identifying the globally optimal solution [26]. By encoding combinatorial problems into Ising spin models and leveraging quantum effects like tunneling, superposition, and entanglement, quantum annealing has the potential to find high-quality solutions more efficiently than classical approaches [27].

Quantum annealing utilizes quantum tunneling to traverse energy barriers and avoid local minima that classical algorithms can get stuck in. The interactions between qubits in the Hamiltonian enable tunneling through barriers in the energy landscape. Quantum fluctuations induced by a transverse field are gradually reduced to control the Hamiltonian's evolution from an initial state to the problem state. Quantum annealing is well-suited for quadratic binary optimization problems and equivalent Ising models [28–30]. Many optimization problems can be mapped to this form and solved on quantum annealing machines with over 5000 qubits now available in commercial systems such as D-Wave. By leveraging quantum effects, it provides a distinct approach to solving complex combinatorial optimization problems.

The quantum annealing method performs well in combinatorial optimization problems, such as the Traveling Salesman Problem and the Graph Coloring Problem [31]. Research on the quantum annealing algorithm encompasses algorithmic improvements, the expansion of its application domains, and integration with other computational methods [32]. It is possible to control the evolution of the system by introducing quantum gate operations to explore the solution space more efficiently [33]. Quantum annealing algorithm has also been applied to geophysical inversion, distribution network fault location, and other fields, showing its potential in practical applications [34,35].

Despite its theoretical promise, practical deployment faces significant energy efficiency limitations and scalability constraints. Current quantum annealers (e.g., D-Wave systems) require cryogenic environments near absolute zero (operating at ~15 mK), maintained by large-scale dilution refrigerators that dominate the system's energy footprint, dwarfing computational energy itself. [36].

Additionally, such systems struggle with problem mapping overhead, qubit connectivity limitations, and noise-induced decoherence. Decoherence is exacerbated by thermal fluctuations from room-temperature control electronics, necessitating complex noise suppression strategies like cryogenic filtering and superconducting interconnects. Consequently, recent research prioritizes hybrid quantum-classical frameworks that integrate quantum annealing with classical preprocessing or post-processing [37,38]. These frameworks delegate sub-tasks to classical hardware, minimizing quantum processor runtime and thereby reducing cryogenic load, a critical mitigation for energy-intensive operations.

The practical viability of quantum annealing systems is therefore closely tied to continued advancements in hardware design, particularly those aimed at reducing the energy consumption of cryogenic infrastructure and control subsystems. In parallel, progress in algorithmic design plays a pivotal role in maximizing the computational value derived from quantum resources. Realizing the full potential of quantum annealing thus depends on coordinated improvements in both physical hardware and computational frameworks. The present study contributes to the algorithmic front by introducing an optimized approach tailored to the constraints of educational exam scheduling.

3.2. Quadratic unconstrained binary optimization

The Quadratic Unconstrained Binary Optimization (QUBO) model plays a pivotal role in formulating combinatorial optimization problems to be solved by quantum annealing [28–30]. QUBO originated from quadratic programming, which allows modeling binary optimization problems with a quadratic objective function and linear constraints. Ising models represent the energy state of atomic spins in a lattice, which quantum annealing aims to minimize. The binary variables in QUBO can be mapped to spin states, with the quadratic coefficients corresponding to coupling strengths between the spins. This establishes a precise equivalence between QUBO and Ising models, serving as a primary approach for quantum computing to solve combinatorial optimization problems [39,40].

The general expression of the QUBO model is shown as Formula (1). In this formula, N is the number of binary variables in the problem, x_i represents the binary value (0 or 1) of the i-th variable, p_i is the linear coefficient, q_{ij} is the quadratic coefficient. And $H(x)$ is the objective function in terms of binary variables, describing the total energy of the problem.

$$H(x) = \sum_{i=1}^{N} p_i x_i + \sum_{i=1}^{N} \sum_{j=i+1}^{N} q_{ij} x_i x_j \tag{1}$$

Once the constrained combinatorial optimization model is transformed into the general QUBO expression as in Formula (1), it becomes amenable to solution using quantum annealing algorithms on quantum computers. A critical step in leveraging quantum computing for solving real-world problems is to effectively formulate the problem as a QUBO model, which entails encoding its objectives and constraints into the Hamiltonian explored by the quantum annealer.

4. Examination scheduling problem description

The examination scheduling issue necessitates the arrangement of test timings for all students within a specified timeframe, as a goal-oriented optimization challenge with multiple constraint conditions.

Based on the actual circumstances at our university, approximately 1807 students are required to take supplementary examinations for a total of 215 courses, owing to the impact of the 2022 pandemic. This accounts for a total of around 2749 exam instances. Among these, some students need to take only a single supplementary examination, while others need to take up to 8. It is crucial to avoid any scheduling conflicts in these students' examination timetables. Additionally, it is essential to ensure that all students enrolled in the same course take their exams simultaneously. According to the university directives, it is imperative to complete all examinations within 3 days, with the possibility of scheduling up to 3 examination sessions per day.

To facilitate the description and illustration of its mathematical model, a small dataset comprising 6 students (S1-S6) taking exams for 10 courses (A-J) is adopted as an illustrative example to explain the specific mathematical model, as shown in Table 1.

To address the exam scheduling problem in this study, the following constraints must be satisfied based on the actual requirements of the scenario:

Upper Bound on Examination Batches: To ensure timely scheduling of exams, different courses are grouped into batches, with all courses within each batch being simultaneously examined. In this example, all students are required to complete all their exams within 5 batches.

Unison of Course Examination Timing: A fundamental requirement dictates that all students enrolled in a particular course must take the examination at the same time.

Exclusivity in Examination Attendance: An intrinsic constraint

Table 1
Simplified Student Schedule.

Student	Courses
S1	A, B, C, D, E
S2	B, C, E, F
S3	A, D, G
S4	A, H, I
S5	C, J
S6	B, J

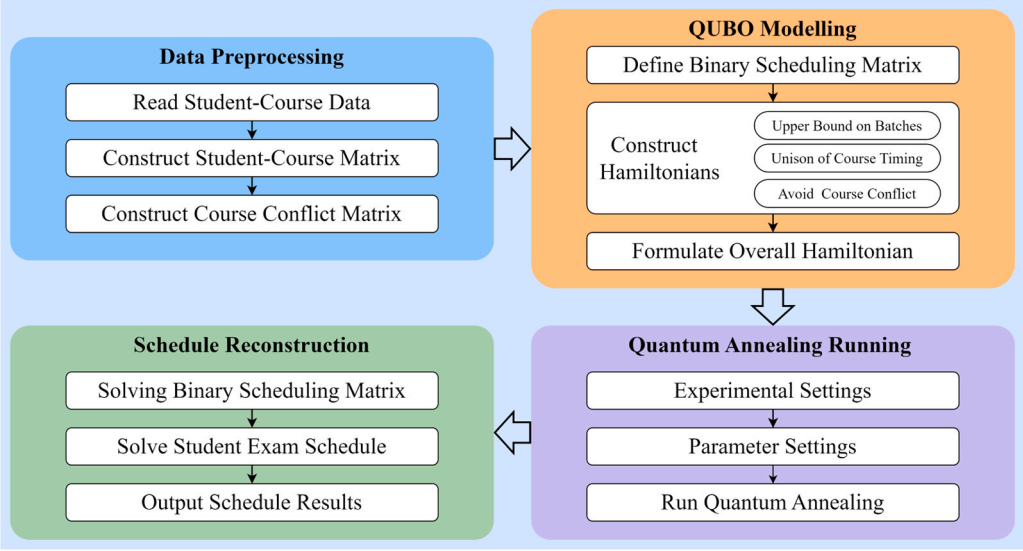


Fig. 1. Quantum Annealing Exam Scheduling Algorithm Flowchart.

mandates that no student can participate in the examination of multiple distinct courses simultaneously.

5. Method

5.1. Algorithm overview

The proposed quantum annealing approach for large-scale exam scheduling consists of four main processing modules, as illustrated in Fig. 1. The algorithm begins with data preprocessing to transform raw student-course registration data into structured matrices suitable for optimization. Subsequently, the QUBO modeling module formulates the scheduling problem as a Hamiltonian optimization problem by defining constraints and objective functions. The quantum annealing module then leverages D-Wave quantum computing systems to solve the optimization problem efficiently. Finally, the schedule reconstruction module converts the quantum solution back into human-readable exam schedules.

To facilitate a clearer understanding of the mathematical formulation and optimization process employed in this study, a symbolic description of the key variables and parameters is systematically presented in Table 2.

5.2. Data preprocessing

The list of make-up exams output by the school's academic affairs office is usually presented in Table 1. To enable effective optimization using quantum annealing, this data needs to be preprocessed.

Specifically, the original data is converted into the "Student ID-Course ID" Matrix A as shown in Fig. 2. $A_{i,j}$ indicates whether the Student i is taking the exam for the Course j.

$$A_{i,j} = \begin{cases} 1, \&\text{Taking the exam.} \\ 0, \&\textit{Not}\text{taking the exam.} \end{cases} \tag{2}$$

Following this transformation, and as detailed in Algorithm 1, the matrix A is converted to the course conflict matrix C to detect examination scheduling conflicts. The algorithm processes each student's examination schedule, identifying conflicts between courses scheduled for the same examination period.

Algorithm 1. Solving the Course Conflict Matrix

```
Input:
  S            // Total number of students
  N            // Total number of courses
  A[S × N]     // Student-course enrollment matrix
Output:
  C[N × N]     // Course conflict matrix
Initialize an undirected graph G with N nodes
for k ← 1 to S do
    for i ← 1 to N do
        for j ← i + 1 to N do
            if A[k, i] = 1 and A[k, j] = 1 then
                G.add_edge(i, j)
            end if
        end for
    end for
end for
C ← G.to_matrix()
```

Table 2
Symbolic description.

Symbol	Description
H	Hamiltonian, the total energy function of a physical system.
N	Total number of courses.
M	Total number of exam batches.
S_i	The number of students selected for the Course i.
n_k	The number of students in the Batch k.
τ_i	The number of exams scheduled for the Course i.

Student \ Course	A	B	C	D	E	F	G	H	I	J
S1	1	1	1	1	1	0	0	0	0	0
S2	0	1	1	0	1	1	0	0	0	0
S3	1	0	0	1	0	0	1	0	0	0
S4	1	0	0	0	0	0	0	1	1	0
S5	0	0	1	0	0	0	0	0	0	1
S6	0	1	0	0	0	0	0	0	0	1

Fig. 2. The "Student ID - Course ID" Matrix.

These examination conflicts are represented in an undirected graph G shown in Fig. 3, where nodes denote courses and edges connect courses that cannot be scheduled simultaneously due to student examination conflicts.

The course conflict matrix C is defined as:

$$C_{i,j} = \begin{cases} 1, \&Conflict \\ 0, \&No\ Conflict \end{cases} \tag{3}$$

The undirected graph G is then converted into the matrix C, as shown in Fig. 4.

After the course conflict matrix is solved, the examination of the same course will not be arranged multiple times, which naturally satisfies the constraint of the consistency of the course examination time.

5.3. QUBO modelling

In this section, the scheduling constraints are translated into a Hamiltonian, which serves as the objective function to be minimized during the quantum annealing process. Unlike many prior QUBO scheduling models that directly encode constraints without an intermediate structure, our approach incorporates a graph-based preprocessing stage. This allows for a more systematic and compact representation of conflict constraints, effectively reducing the complexity of the Hamiltonian.

The binary qubit matrix X is defined as the exam schedule, where $X_{i,k}$

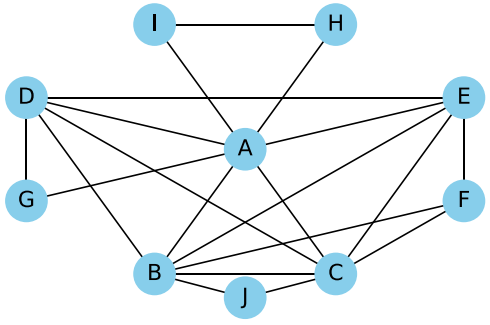


Fig. 3. The undirected graph G.

Course i \	A	B	C	D	E	F	G	H	I	J
A	0	1	1	1	1	0	1	1	1	0
B	1	0	1	1	1	1	0	0	0	1
C	1	1	0	1	1	1	0	0	0	1
D	1	1	1	0	1	0	1	0	0	0
E	1	1	1	1	0	1	0	0	0	0
F	0	1	1	0	1	0	0	0	0	0
G	1	0	0	1	0	0	0	0	0	0
H	1	0	0	0	0	0	0	0	1	0
I	1	0	0	0	0	0	0	1	0	0
J	0	1	1	0	0	0	0	0	0	0

Fig. 4. The Course Conflict Matrix.

indicates whether the Course i is tested in Batch k, and $i \in [1, N], j \in [1, M]$.

$$X_{i,k} = \begin{cases} 1, \&Schedule\ Examination. \\ 0, \&Schedule\ No\ Examination. \end{cases} \tag{4}$$

The upper bound on examination batches is set by defining only k batches in the binary matrix X, ensuring that all exams can be scheduled within the allotted time.

To ensure each course is scheduled for exactly one exam, the number of exams for Course i is denoted as τ_i and require that:

$$\tau_i = 1, \tag{5}$$

where τ_i is calculated as follows:

$$\tau_i = \sum_{k=1}^{M} X_{i,k}, \tag{6}$$

where k represents the index of the batch, and M denotes the total number of batches. The constraint can be converted to Hamiltonian as shown in the following formula:

$$H_{const1} = \sum_{i=1}^{N} (\tau_i - 1)^2 \tag{7}$$

To avoid conflicting courses being scheduled in the same batch of examination, $C_{i,j},\ X_{i,k},\ X_{j,k}$ should not be 1 at the same time. Both C and X are binary matrices, so the constraint can be converted to the following formula:

$$C_{i,j} X_{i,k} X_{j,k} = 0 \tag{8}$$

Then, the constraint is converted to Hamiltonian as shown in the following formula:

$$H_{const2} = \sum_{i=1}^{N} \sum_{j=1}^{N} \sum_{k=1}^{M} C_{i,j} X_{i,k} X_{j,k} \tag{9}$$

By combining Hamilton of the constraints and the objective, the complex scheduling problem is modeled as an unconstrained binary optimization problem.

$$H = \lambda H_{const1} + \gamma H_{const2} \tag{10}$$

Subsequently, we can use the quantum annealing algorithm to find the solution for the binary matrix X. When H reaches zero, indicating that all constraints are satisfied, we will have obtained a feasible examination scheduling solution.

Course	1	2	3	4	5
A	0	0	0	1	0
B	0	1	0	0	0
C	0	0	0	0	1
D	0	0	1	0	0
E	1	0	0	0	0
F	0	0	1	0	0
G	0	1	0	0	0
H	0	0	1	0	0
I	0	0	0	0	1
J	1	0	0	0	0

Fig. 5. The binary matrix X of exam scheduling solved by a quantum computer.

5.4. Quantum annealing running

The quantum annealing algorithm runs on the D-Wave Advantage System 6.4 quantum processor, which provides 5612 qubits for the quantum annealing process. The Leap Hybrid Sampler uses the hybrid binary quadratic model version 2. The classical simulated annealing algorithm runs on an Intel i5 10400 CPU. These settings provide the basis for comparing the performance of the algorithms subsequently.

For implementing the QUBO model, the PyQUBO library[41] in Python was used to automatically parse, validate, and compile the QUBO formulation into the Binary Quadratic Model (BQM). This format is suitable for input into various solvers, including D-Wave's quantum annealing solvers and simulated annealing solvers. Although the model in this research contained only quadratic terms and PyQUBO was not strictly required, its flexibility was advantageous in simplifying the process of building and managing QUBO expressions, enhancing modularity, and streamlining the workflow. Additionally, by producing solver-agnostic outputs, PyQUBO facilitates easy adaptation to different solvers, thereby improving overall efficiency and reproducibility.

In the quantum annealing experiments, the number of reads was set to 500, and the annealing time was set to 200 microseconds. To handle the minor embedding of the problem, we used the D-Wave Embedding Composite tool, which integrates with the 'minorminer' library for the embedding process. This tool automatically maps the binary quadratic model onto the quantum processor's hardware. For chain break resolution, we applied the majority vote algorithm, ensuring consistent decisions when chains representing logical qubits break during annealing.

In the hybrid annealing approach, all parameters retained their default values, and no special parameters were specified. This demonstrates the solver's general applicability and effectiveness under standard conditions, showcasing its ability to handle a wide range of problems with the default settings provided.

For the simulated annealing process, we set the number of reads to 50 and the number of sweeps to 10,000. The number of reads specifies how many times the algorithm will execute the annealing process to sample different solutions, while the number of sweeps determines the total number of sweeps or iterations performed during each run.

5.5. Schedule reconstruction

Following the completion of the quantum annealing process, the solution must be translated into a structured exam timetable suitable for operational use. This reconstruction consists of two primary stages:

The first step is to interpret the binary scheduling matrix $X\epsilon\{0,1\}^{N\times M}$, which is produced by the quantum solver and represents the batch assignment for each course. Each row corresponds to a course, and each column corresponds to an exam batch. An element $x_{ij} = 1$ indicates that the Course i is assigned to the Batch j. The constraint formulation ensures that each course is assigned to exactly one batch. Since this matrix satisfies all imposed constraints, it can be interpreted without further transformation. For validation and administrative review, the matrix is presented in tabular form, in which scheduled entries are explicitly identified.

The second step is to generate the student exam schedule. Based on the batch assignments in the matrix X and the enrollment matrix $A\epsilon\{0,1\}^{S\times N}$, where S is the number of students, a student-batch matrix $S_{plan\epsilon}T^{S\times M}$ is constructed. For each course assigned to a specific batch, all students registered in that course are determined from the matrix A, and their corresponding entries in S_{plan} are updated with the course identifier. The detailed procedure is outlined in Algorithm 2. Each row of S_{plan} thus reflects the complete exam allocation for a student across all batches. To support schedule verification and deployment, the matrix S_{plan} is visualized using structured tables.

Algorithm 2. Student Exam Schedule Generation

```
Input:
  X[N × M]      // Course-to-batch scheduling matrix
  A[S × N]      // Student-course enrollment matrix
Output:
  Splan[S × M]  // Student-specific exam schedule
Initialize Splan[S × M] ← "-"
for i ← 1 to N do
  for j ← 1 to M do
    if X[i, j] = 1 then
      for s ← 1 to S do
        if A[s, i] = 1 then
          Splan[s, j] ← i
        end if
      end for
    end if
  end for
end for
```

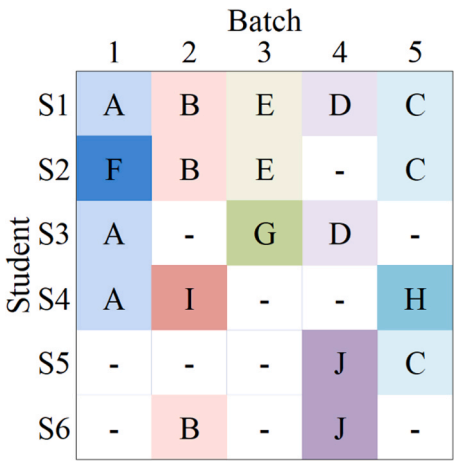


Fig. 6. The student exam schedule.

Table 3
Performance Metrics Across Different Problem Sizes.

Number of examinations	19	1000	1500	2000	2749
Number of students	6	850	1206	1442	1807
Number of courses	10	27	72	129	215
Min batches	5	5	6	7	8
Number of variables	50	135	432	903	1720
Number of quadratic terms	250	565	2154	5488	13348
QPU execution time of native	149 ms	172 ms	192 ms[a]	-[b]	-[b]
QPU execution time of hybrid	74 ms	81 ms	84 ms	85 ms	86 ms
CPU execution time of simulated	346 ms	939 ms	3319 ms	6547 ms	13547 ms
CPU execution time of Gurobi	25 ms	58 ms	60 ms	75 ms	153 ms

[a] For 1500 examinations, quantum computing can produce results within 192 ms, but the solutions are discarded as they fail to meet the constraint requirements.

[b] When the number of examinations exceeds 1500, the embedding process becomes increasingly complex due to the substantial increase in both variables and quadratic terms, which ultimately prevents the effective execution of the quantum annealing algorithm.

6. Result

6.1. Schedule reconstruction and interpretation

The output obtained from the hybrid quantum annealing process provides a complete, conflict-free scheduling solution that satisfies all predefined constraints. As illustrated in Fig. 5, the binary scheduling matrix accurately assigns each exam course to one of the available time batches. Each course appears in exactly one batch, and no two conflicting courses are scheduled simultaneously. This verifies the global feasibility of the quantum-optimized solution for both temporal exclusivity and course-level synchronization.

Based on this batch-level assignment, student-specific timetables are derived by mapping the scheduled courses to the corresponding students. The resulting matrix, presented in Fig. 6, offers a concise and structured overview of the individual exam arrangements. Each row represents a student, and each column corresponds to an exam batch. The cell entries specify which exam, if any, the student is scheduled to take during that batch.

The final schedule exhibits complete coverage: all required exams are accommodated within the predefined batch limits, and no student is assigned multiple exams in the same time slot. This demonstrates the practical soundness and operational applicability of the solution.

6.2. Performance metrics across different problem sizes

To evaluate the scalability and performance of different annealing algorithms, we conducted experiments across varying problem sizes, ranging from a simplified 19-exam instance to the full real-world dataset containing 2749 exam instances. Table 3 summarizes key performance metrics, including the number of students, courses, binary variables, and quadratic terms involved in each instance. As problem size increases, simulated annealing shows a steep rise in CPU execution time, highlighting its computational inefficiency at scale. In contrast, quantum annealing maintains relatively low QPU execution time for small to medium-sized problems but fails to yield valid solutions beyond 1500 exams due to embedding limitations on current hardware. Notably, hybrid quantum annealing achieves stable QPU performance—ranging from 74 ms to 86ms—across all problem sizes, demonstrating both efficiency and robustness. This trend is further illustrated in Fig. 7, which provides a visual comparison of solver execution time under increasing problem complexity.

Simulated annealing, while effective at smaller scales, shows a significant increase in CPU execution time as the problem size grows, indicating its limitations in handling larger problem instances. Quantum annealing shows considerable promise, particularly for smaller-scale problems, with QPU execution time between 149 and 192 ms. However, the limitations of quantum annealing become evident as the problem size increases, primarily due to the increasing difficulty of embedding. In contrast, hybrid annealing consistently delivers low QPU execution time across all scales, from 74 to 86 ms, effectively balancing quantum and classical resources while maintaining stable execution time. This highlights the potential of hybrid approaches to handle increasingly complex problems more efficiently than either classical or quantum methods alone.

Apart from the simulated annealing algorithm, the QUBO formulas in this paper were input into the Gurobi solver for evaluation [42]. Gurobi is widely regarded as one of the fastest solvers available today. It combines advanced techniques such as preprocessing, branch-and-bound, and multithreading. These methods make it highly efficient for solving various problems, including linear programming, mixed-integer linear programming, quadratic programming, and constraint programming. Notably, quantum annealing achieved comparable performance to Gurobi, as shown in Table 3, with a slight speed advantage. While the performance edge was not dramatic, it is encouraging to see quantum annealing being competitive with a leading classical solver like Gurobi. As quantum hardware continues to advance and scale, we expect this advantage to become more significant.

7. Discussion

The simulated annealing algorithm simulates the physical annealing process, by randomly selecting the initial solution, and accepting or rejecting the new solution according to a certain probability in each iteration, to gradually approach the global optimal solution. This process involves a gradual reduction of temperature parameters, as well as enough iterations at each temperature to ensure that the solution space is fully explored. It is straightforward to implement on classical hardware and quite flexible in encoding optimization problems [43].

Although the simulated annealing algorithm is simple and easy to implement, it is usually less efficient than the quantum annealing algorithm when dealing with large-scale problems. Quantum annealing algorithm can make use of quantum parallelism and quantum interference effect to significantly improve the search efficiency [44]. Compared with the simulated annealing algorithm, the time complexity

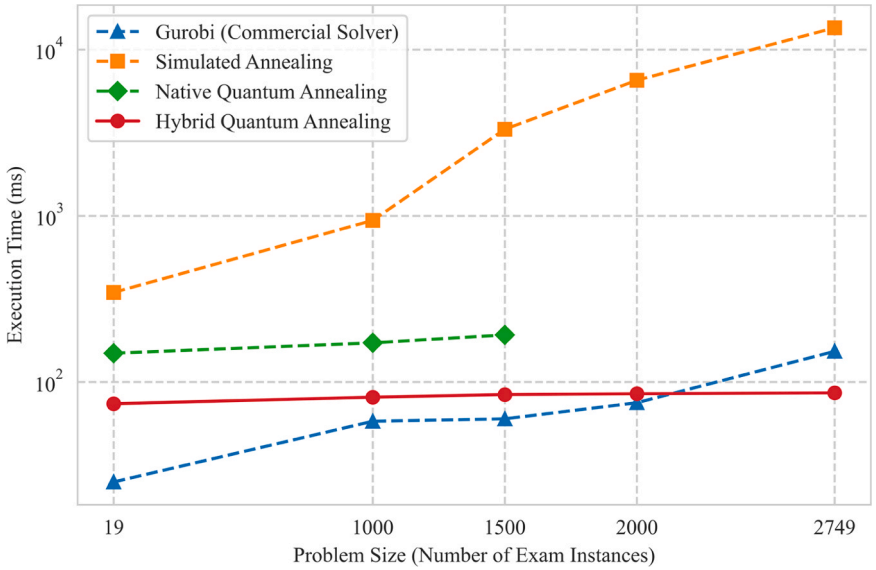


Fig. 7. Execution Time Comparison of Different Solvers Across Problem Sizes.

Table 4
Comparison of different annealing methods.

Method	Quantum	Hybrid	Simulated
Speed	High	Moderate	Slow
Core	QPU	Hybrid	CPU
Working Qubits	5612	Flexible	0
Flexibility	Fixed	Flexible	Flexible
Operating Environment	Stringent	Stringent	Lenient
Scalability	Limited	Moderate	High
Quantum Environment Noise	Include	Include	None
Optimization Quality	Specialized	Balanced	General

is lower because a lot of the iterative process is eliminated. When the number of courses to schedule is relatively small, the native quantum annealing on the D-Wave QPU was utilized to provide true quantum acceleration by exploiting quantum effects like tunneling and superposition. The quantum advantage manifests in finding quality solutions much faster than classical algorithms.

However, the native quantum annealing is limited by the number of available qubits on current quantum hardware. When scheduling a large number of courses that exceed the qubit limits of the QPU, the hybrid quantum annealing algorithm, which combines quantum annealing and classical algorithms, can be a better choice. The hybrid quantum annealing algorithm overcomes the limited qubits on the D-Wave quantum processing unit by seamlessly combining quantum annealing executed on quantum hardware with optimization algorithms run on classical solvers. This hybrid quantum annealing algorithm provides the flexibility to handle large problem sizes exceeding the capabilities of current quantum devices. By utilizing both quantum and classical resources, it solves very large problems that current QPUs cannot accommodate. For large and complex problems, the hybrid sampler leverages quantum tunneling effects to escape local optima, while inheriting the precision and scalability of classical algorithms [45].

As summarized in Table 4, each annealing sampler has complementary advantages suited for particular problem scenarios, scales, and hardware constraints. To solve the exam scheduling problem, various solutions are available, catering to different requirements and resources. For small instances, the quantum annealing provides direct acceleration, leveraging quantum effects. For large-scale scheduling, hybrid quantum annealing combines the strengths of quantum and classical algorithms to achieve flexible and scalable performance. Simulated annealing provides a classical alternative when quantum resources are unavailable. By understanding these trade-offs, an appropriate approach can be selected to optimize exam scheduling for the specific needs and resources at hand.

There are still many possibilities to explore for enhancing the exam scheduling model and improving the performance of quantum annealing algorithms. Future work may investigate more parameters in the model construction, such as taking into account the seating capacity and features of different exam rooms. The performance evaluation can also be expanded by testing on more exam scheduling scenarios and comparing with more classical optimization algorithms. Quantum computing provides a new solution for large-scale, multi-scenario scheduling problems in the future.

8. Conclusions

This study applied hybrid quantum annealing to address the complex problem of large-scale exam scheduling in real-world educational scenarios, particularly during makeup and graduation examinations involving 2749 exam instances. A key contribution lies in integrating classical graph-based preprocessing with QUBO model construction, which streamlines constraint encoding and enables quantum solvers to scale to problem sizes beyond current hardware limitations. The proposed method successfully addressed multiple constraints, including student-course conflicts, room limitations, and batch synchronization. The results demonstrated that hybrid quantum annealing achieved stable performance with a QPU execution time of 86 ms, significantly outperforming classical simulated annealing by over two orders of magnitude in speed. This establishes a viable pathway for quantum-enhanced scheduling solutions.

Future research could explore the integration of additional practical constraints, such as classroom capacity limits, subject-specific exam duration variability, and strategies to minimize consecutive exams for individual students. Expanding the model's applicability to other complex timetabling scenarios would further demonstrate its generalizability. Additionally, conducting broader benchmarks against a wide range of hybrid metaheuristics and commercial solvers would help to more comprehensively validate the robustness and computational efficiency of the proposed approach. With ongoing advances in quantum computing hardware, hybrid frameworks such as the one presented in this study are expected to find broader applications in educational scheduling and other complex optimization domains.

CRediT authorship contribution statement

Ziyu Zhou: Writing – original draft. **Shuyan Li:** Writing – review & editing, Project administration. **Mengtong Tan:** Validation. **Chaojie Zhang:** Writing – review & editing, Conceptualization. **Qing Chen:** Data curation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used multiple generative AI tools to refine the linguistic aspects of the manuscript only. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of Competing Interest

The authors report there are no competing interests to declare.

Data availability

The code and all data (anonymized) used in this research can be accessed at https://github.com/zhouziyu2050/QuantumExamScheduling.

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